INCH-POUND

MIL-DTL-28875/5B
20 January 2015
SUPERSEDING
MIL-DTL-28875/5A
24 August 2000

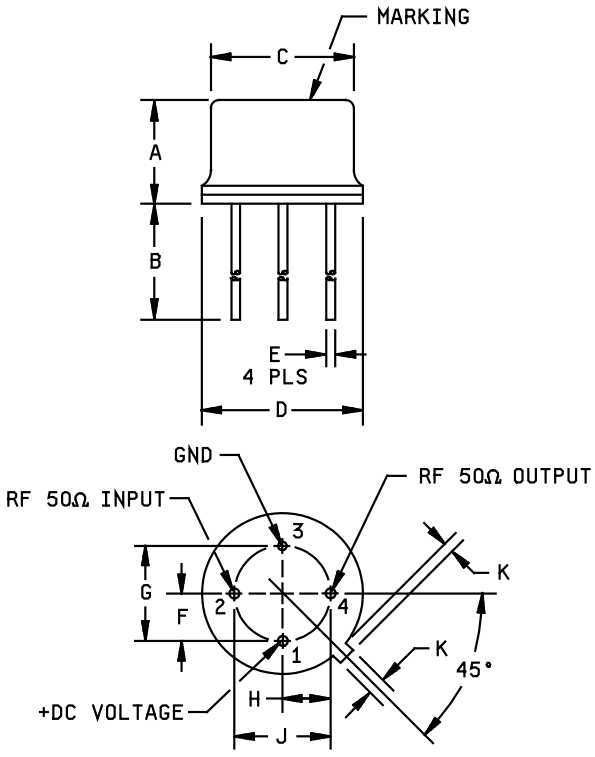
DETAIL SPECIFICATION SHEET

AMPLIFIERS, RADIO-FREQUENCY AND MICROWAVE, SOLID-STATE, LOW NOISE, WIDEBAND, TO CONFIGURATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Inactive for new design
after 20 May 2010

The requirements for acquiring the amplifiers described herein shall consist of this specification sheet and MIL-DTL-28875.



	Inches		Millimeters	
	Max	Min	Max	Min
B	N/A	.17	N/A	4.3
C	.455 dia	.445 dia	11.56 dia	11.30 dia
D	.510	.490	12.95	12.45
E	.019	.016	.48	.41
F	.160	.140	4.06	3.56
G	.310	.290	7.87	7.37
H	.160	.140	4.06	3.56
J	.310	.290	7.87	7.37
K	.05	N/A	1.3	N/A

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only (1.00 in = 25.4 mm).

FIGURE 1. Dimensions and configuration.

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TABLE I. Electrical characteristics (25°C unless otherwise specified).

Dash No.	Frequency range (MHz)	DC input power		Gain (dB)	Gain flatness (dB)	RF output power (dBm)	VSWR Max		Noise figure (dB)	Intercept point 3rd order (dBm)	Harmonic output (dBc)	Gain versus temp. (dB)	RF input power (dBm)	Isolation (dB)	Phase Linearity (deg)
		Voltage	Curr. Max (mA)	Min	Max	Min	IN	OUT	Max	Min			Max	Min	Max
S1 01 S2 N	5-250	+15 ±1%	40	31 29 from -54°/+85°C	±0.2 1.0 from -54°/+85°C	+9 +7 from -54°/+85°C	1.3:1 2.0:1 from +85°C	1.3:1 2.0:1 -54°C	4 5.5 from -54°/+85°C	+21	N/A	±2	+17	N/A	N/A
S1 02 S2 N	200-400	+15 V dc ±1 V dc	20	12.2	±0.5	+6	1.17:1	1.17:1	2.5	+19	N/A	N/A	+13	25	±.5
S1 03 S2 N	5-500	+15 ±1%	25	20	±1.0	+7	2.0:1	2.0:1	4.5	+20	N/A	N/A	N/A	N/A	N/A
S1 04 S2 N	5-500	+15 ±1%	10	15	±1.0	-2 -3 from -54/+85°C	2.0:1	2.0:1	2.5 3.0 from -54/+85°C	+8	N/A	N/A	+13	N/A	N/A
S1 05 S2 N	5-500	+15 ±1%	10	14 13.5 from -54/+85°C	±1.0	-2 -3 from -54/+85°C	2.0:1	2.0:1	4 4.5 from -54/+85°C	+11	N/A	±0.5	+13	N/A	N/A
S1 06 S2 N	5-500	+15 ±1%	23	20 19 from -54/+85°C	±1.0	+7	2.0:1	2.0:1	4.5 5.0 from -54/+85°C	+20	N/A	±1.0	+13	N/A	N/A
S1 07 S2 N	5-500	+15 ±1%	23	14 13.5 from -54/+85°C	±1.0	+7	2.0:1	2.0:1	5.5 6.0 from -54/+85°C	+21	N/A	±0.5	+13	N/A	N/A
S1 08 S2 N	2-1000	+15 ±1%	10	14 13.5 from -54/+85°C	±0.7 ±1.0 from -54/+85°C	-5 -6 from -54/+85°C	2.0:1	2.2:1	3.5 4 from -54/+85°C	+10	N/A	N/A	+13	N/A	N/A
S1 09 S2 N	5-1000	+15 ±1%	25	14 13 from -54/+85°C	±1.0 ±1.2 from -54/+85°C	+7 +6.5 from -54/+85°C	2.0:1	2.0:1	6.5 7 from -54/+85°C	+21	N/A	N/A	+17	N/A	N/A

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TABLE I. Electrical characteristics (25°C unless otherwise specified) - Continued.

Dash No.	Frequency range (MHz)	DC input power		Gain (dB)	Gain flatness (dB)	RF output power (dBm)	VSWR Max		Noise figure (dB)	Inter-cept point 3rd order (dBm)	Har-monic output (dBc)	Gain versus temp. (dB)	RF input power (dBm)	Iso-lation (dB)	Phase Linearity (deg)
		Voltage	Curr. Max (mA)	Min	Max	Min	IN	OUT	Max	Min			Max	Min	Max
S1 10 S2 N	10-1200	+15 ±1%	40	24 23 from - 54°/+85°C	±0.8 1.0 from -54°/+85°C	+7 +6.5 from -54°/+85°C	2.0:1	2.0:1	4.3 +4.8 from - 54°/+85°C	+20	N/A	N/A	+17	N/A	N/A
S1 11 S2 N	10-2000	+15 ±1%	12	10.5 from - 54°/+100° C	±1.0 from -54°/+100°C	-4.5 from - 54°/+100°C	2.2:1	2.2:1	4.5 from - 54/+100°C	+10	N/A	N/A	+27	N/A	N/A
S1 12 S2 N	100-2000	+15 ±1%	40	15 14 from - 54°/+85°C	±0.7 +1.0 from -54°/+85°C	+6 +5 from -54°/+85°C	2.0:1	2.0:1	6.5 7.0 from - 54°/+85°C	+18	N/A	N/A	+17	N/A	N/A

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TABLE II. Dimension, physical and environmental characteristics.

Dash number	Dimension A in inches (mm) Max	Mass Max (Grams)	Temperature range		Burn-in temperature (case temp)	Seal	Salt
			Operating case temp	Nonoperating			
01 S1 S2 N	.22 (5,6)	2.5	-54/+85°C	-65/+100°C	+100°C	Hermetic	Applicable
02 S1 S2 N	.27 (6.8)	2.5	-55/+100°C	-65/+125°C	+100°C	Hermetic	Applicable
03 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
04 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+100°C	Hermetic	Applicable
05 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
06 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
07 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
08 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
09 S1 S2 N	.18 (4.6)	2.5	-54/+85°C	-65/+100°C	+125°C	Hermetic	Applicable
10 S1 S2 N	.22 (5.6)	2.5	-54/+85°C	-65/+100°C	+100°C	Hermetic	Applicable
11 S1 S2 N	.22 (5.6)	2.5	-54/+100°C	-62/+125°C	+125°C	Hermetic	Applicable
12 S1 S2 N	.22 (5.6)	2.5	-54/+100°C	-62/+125°C	+125°C	Hermetic	Applicable

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table II.

Electrical characteristics: See table I.

Physical and environmental characteristics: See table II.

Part number. M28875/5-(dash number from tables I and II).

Referenced documents. This specification sheet only references MIL-DTL-28875.

The margins of this specification sheet are marked with vertical lines to indicate where modifications from this revision were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodian:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5996-2015-002)

Review activities:
Navy - AS, MC, OS, SH
Air Force - 99

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